

Puppala Madhuvenu

Java Backend Developer — Java — Spring Boot — SQL

Bengaluru, India | puppalamadhuvenu9391@gmail.com | +91-9391808204

github.com/Madhu-9391 | Portfolio | linkedin.com/in/madhuvenu-vits

Professional Summary

B.Tech Computer Science Graduate (2026) with **CGPA 8.99/10** and hands-on experience developing **Java, Spring Boot, SQL, MySQL, and RESTful API** applications. Strong foundation in **Core Java, OOP, Collections, exception handling, data structures, and modular application design**, with practical experience debugging backend services, optimizing SQL/database access, and integrating React.js frontends with REST APIs. Built and deployed backend systems with measurable improvements in **API latency, throughput, and reliability**, demonstrating a practical, problem-solving approach to software development.

Education

PBR Visvodaya Institute of Technology and Science, Kavali
B.Tech in Computer Science and Engineering

2022 – 2026
CGPA: 8.99/10

Technical Skills

Programming: Java, Python, JavaScript, SQL

Java: Core Java, OOP, Collections, Exception Handling, Java 8+ Concepts

Backend: Spring Boot, Spring Security, JPA/Hibernate, RESTful APIs

Frontend: React.js, HTML5, CSS3, WebSocket

Databases: MySQL, Relational Database Design, SQL Query Optimization

Software Engineering: Data Structures & Algorithms, Modular Architecture, SDLC, Debugging, API Development

Tools: Git, GitHub, Maven, Postman, VS Code, Docker

Projects

AI-Powered Intelligent Smart Queue Management System **Live Demo** **GitHub** *May 2026 – Jul 2026*

- Designed and developed a **Java Spring Boot backend** with REST APIs, MySQL, JPA/Hibernate, and WebSocket, structuring appointment, token, queue, analytics, and notification workflows into reusable backend components.
- Implemented backend business logic for **appointment booking, token generation, queue management, ETA recalculation, analytics, and role-based workflows**, applying OOP and modular service-layer design to keep functionality maintainable.
- Optimized Spring Security authentication by eliminating redundant database lookups and reusing authenticated user context, reducing login latency by **44% (498 ms → 280 ms)**, p95 latency by **48% (1.52 s → 793 ms)**, and increasing throughput by **17% (30 → 35 req/s)** under **100 concurrent users**.
- Debugged and optimized database access by eliminating **N+1 queries and unnecessary entity loading**, reducing analytics API p95 latency by **45% (1.30 s → 711 ms)** and average latency by **58% (419 ms → 177 ms)** at **1,000 concurrent users**.
- Validated REST API reliability across authentication, appointments, queue management, analytics, and doctor workflows, sustaining **43.7 req/s, 26 ms average latency, 98.6 ms p95 latency, and 0% request failures** under **100 concurrent users** using k6.

Enterprise Employee Management Platform – Java Spring Boot **Live Demo** **GitHub** *Jan 2025 – Mar 2025*

- Developed a full-stack application using **Java, Spring Boot, Spring Security, Hibernate/JPA, MySQL, REST APIs, and React.js**, integrating frontend workflows with backend APIs for employee, department, document, and dashboard operations.
- Structured complex employee-management business logic into **reusable controller, service, and repository components**, applying OOP, JPA entity relationships, validation, and transactional service-layer operations.
- Implemented **JWT-based authentication and role-based authorization** for ADMIN, HR, and MANAGER users using Spring Security, securing REST endpoints with access and refresh token handling.
- Built REST APIs with **pagination, filtering, sorting, JPA Specifications, global exception handling, and audit logging**, improving API maintainability and avoiding inefficient retrieval of large employee datasets.
- Packaged the application using **Docker** and maintained source code through Git/GitHub, supporting repeatable application setup and deployment.

Professional Experience

CodeTantra

May 2025 – Jun 2025

Java Full Stack Intern **Certificate**

- Developed hotel-booking application workflows using **Java, Spring Boot, MySQL, REST APIs, and persistence logic**, translating functional requirements into backend application components.
- Debugged slow hotel-booking requests and optimized database operations and Spring Boot service-layer processing, reducing API latency by **25% (400 ms → 300 ms)**.